

PilotFish Sandbox — Interface Construction Playbook

PilotFish Sandbox — Interface Construction Playbook

Audience: Cursor agents (and humans reviewing agent work).
Scope: Any new or substantially changed PilotFish interface under Clients/.
Purpose: One repeatable way to design, scaffold, implement, document, and smoke-test interfaces so demos stay consistent and production risks are called out honestly.

Do **not** invent a one-off layout. Follow this playbook, then specialize only where the integration requires it.

How agents know how to build PilotFish interfaces

Cursor agents are **not** formally trained or certified by PilotFish. There is no special PilotFish fine-tune behind this Sandbox. Knowledge comes from a stack of sources when building an interface here:

1. Authoritative sources in this workspace

Priority	Source	Use it for
A	PilotFish_Documentation/	Module inventory , UI-type → class mapping, config semantics, deep-dive behavior (see §1.1)
A	PilotFish_V2/ (XCS / module Java + modules.conf)	Same EIP module code V1 uses — FQCNs, ConfigurationItem names/defaults, listener/processor/transport behavior (see §1.2)
B	Working Sandbox demos under Clients/**	Proven wiring of those modules into V1 route.xml on pilotfish-eip:23R1
B	This playbook + .cursor/rules/	Construction process, V1 vs V2 policy, risks, definition of done
C	Your chat + DESIGN.md	Job-specific contracts and overrides
D	Live smoke tests / logs/eip.log	Confirm the chosen module loads on the demo image
E	General model knowledge	Only to fill gaps after A–D

Important: V2 does **not** invent a separate module stack. EIP modules (Listeners / Processors / Transports / Routing) are essentially the **same Java modules** whether the route document is classic V1 route.xml or V2 route.v2.xml. V2 mainly changes **how modules are connected and laid out**. Therefore agents **must use PilotFish_V2 and PilotFish_Documentation to decide which modules exist and how they are configured**, then still **author runtime routes as V1** for this Sandbox’s eiPlatform image unless the user overrides.

1.1 PilotFish_Documentation/ (deep-dive library)

Path	Role
README.md	Layout: Listeners / Processors / Transports × version (XCS, 23R1.127, 26R1.11, ...)
General/Process/PilotFish-Module-Documentation-Tracker.md (+ PDF if present)	Master inventory — UI type, short class name, which versions have deep dives
Listeners/, Processors/, Transports/ (<version>/...)	Deep-dive config reference for that module
General/XCS/*	Framework (threads, transactions) when designing concurrency / TX behavior
General/Process/*	How docs were generated — rarely needed for interface build

How to use when constructing an interface:

1. Open the **tracker** and find the UI type you need (e.g. Directory / File, HL7 LLP, Database Polling (SQL)).
2. Note the **class** (DirectoryListener, DatabaseSqlListener, ...) and whether a deep-dive exists for a version close to the runtime image (pilotfish-eip:23R1 demos → prefer 23R1.* docs when present; otherwise use 26R1.11 / XCS and call out version skew in DESIGN.md).
3. Read the deep-dive for config item names, defaults, and pitfalls **before** inventing XML.
4. If the tracker lists a module but there is **no** deep-dive yet, fall through to PilotFish_V2 Java / modules.conf (§1.2), then a Sandbox example if any.

Do **not** skip Documentation when a relevant deep-dive exists.

1.2 PilotFish_V2/ (module source + catalog — shared with V1)

Primary tree: PilotFish_V2/pilotfishdevelopment-xcs-b313c66ccd5f/

Path	Role
XCS-console-gui/.../modules.conf	Catalog of loadable module FQCNs
subprojects/modules-*/src/main/java/com/pilotfish/eip/modules/**	Module implementation — package + class = route.xml class="..."
Configuration / descriptor types on those classes	Exact config element names and defaults under <Configuration>
.clinerules/BUEPRINT.md (or .aiassistant/rules/BUEPRINT.md)	V1 stage model vs V2 node model; what conversion should preserve
V2_RouteViewer/	Viewer/reference assets for diagrams

How to use when constructing an interface:

1. Search modules.conf and/or Java under com/pilotfish/eip/modules for the capability (e.g. SNIP, DirectoryListener, HL7TCP).
2. Resolve the **fully qualified class name** from the Java package + class (or the modules.conf line).
3. Read configuration item definitions (and nearby tests/resources) so XML tags match the code — **do not invent tag names**.
4. Prefer modules that also appear in Sandbox demos when you need a drop-in XML skeleton; otherwise scaffold from V2 + Documentation, then smoke-test on pilotfish-eip:23R1.
5. Use BLUEPRINT / importer notes when mapping V1 Source/Target/Router into generated route.v2.xml.

Wrong package / class strings have already broken demos. Treat FQCNs as **looked-up constants** from Documentation/V2 (or a working demo), never as creative writing.

1.3 Sandbox demos (Clients/**) — proven assemblies

Best source for **end-to-end patterns** that already run in Docker:

Demo	Pattern
Clients/Demos/edi-837-snip-sqlserver/	SQL poll → EIP handoff → fork → EDI + SNIP
Clients/Demos/hl7-healthcare-automation/	Directory → validate → router fan-out → SQL + file

Default assembly method: choose modules via Documentation + V2 → copy a Sandbox route.xml fragment when one exists for that class → otherwise build V1 XML from looked-up config items → convert to V2 for diagrams.

2. How agents pick processors / class names (route.xml)

Decision order:

1. **Identify the capability** (directory listen, SQL poll, XSLT, SNIP, HL7 LLP, ...).
2. **PilotFish_Documentation tracker** → UI type + class; read deep-dive if present.
3. **PilotFish_V2 modules.conf + Java** → confirm FQCN and configuration item XML names.
4. **Sandbox demo route.xml** → copy a working module block when available.
5. **Smoke test** on the demo EIP image; fix using logs if the class is missing from that WAR (docs/V2 may be newer than 23R1).
6. **Ask the user** if the module isn't in the image and can't be substituted.
7. **General model knowledge** only as a last-resort hint — never the source of an FQCN.

3. V1 vs V2 — runtime vs diagrams

Artifact	Format	Role
pilotfish/demo-eip-root/**/route.xml	V1	Runtime for pilotfish-eip:23R1 — author here
eip-root/**/route.v2.xml + modules/*.xml	V2	Routes tab / docs / PDF — usually generated from V1
tools/convert_routes_to_v2.py	V1 → V2	Keeps route.xml; emits diagram graph
PilotFish_V2/	Product + shared module source	Module catalog & behavior and V2 format/editor understanding
PilotFish_Documentation/	Deep dives + tracker	Interface creation decisions (which module, which settings)

Policy: Use V2 **code** and Documentation to choose and configure modules; still **ship V1 route.xml as runtime** and generate V2 for visualization, unless the user explicitly wants native V2 runtime.

4. Instructions you give in chat

Your prompts, corrections, ports, case-study links, and overrides win over defaults. DESIGN.md is the working spec for that interface.

5. General model knowledge (not PilotFish training)

Horizontal knowledge (integration patterns, HL7/EDI concepts, Docker, SQL, Flask) — **not** a substitute for PilotFish_Documentation or PilotFish_V2 when picking modules.

6. External / fetched material (when used)

Public case studies and links you provide; live smoke evidence. Prefer in-repo Documentation + V2 over stale web pages when they conflict on config item names.

7. What is *not* a source of authority

- Invented FQCNs or config element names
- “Only demos may be used” as a hard ceiling when Documentation/V2 clearly provide a better module (demo copying is preferred for **skeletons**, not an excuse to ignore the catalog)
- Claiming compliance without fail-closed config
- Assuming Documentation 26R1 deep-dives always match pilotfish-eip:23R1 without smoke test
- Hand-maintained V2 as the only runtime truth for current Sandbox demos

Practical rule of thumb

Your chat + DESIGN.md

- PilotFish_Documentation (tracker + deep dives)
 - PilotFish_V2 (modules.conf + module Java = same modules as V1)
 - nearest Sandbox demo route.xml (assembly skeleton)
 - author V1 runtime → convert to V2 for viewer/PDF
 - smoke test on pilotfish-eip:23R1

If those disagree, **prefer V2/docs for module identity + Sandbox/live smoke for “loads on this image”**, and document version skew in DESIGN.md.

0. When this applies

Use this playbook when asked to:

- Create a new PilotFish demo / interface / route stack
- Port a case study into a runnable Sandbox demo
- Add a route, listener, transform, or transport to an existing Sandbox interface
- Produce a design document for an interface

If the task is a tiny fix (typo, port conflict, one SQL row), skip the full scaffold and only apply relevant sections (smoke test, honesty about validation, no secret sprawl).

1. Pre-flight (block implementation until answered)

Capture answers in the interface DESIGN.md (see §8). If missing, ask the user.

#	Question	Why
1	Business job in one sentence?	Keeps scope honest
2	Inbound contract (format, envelope, transport: dir / MLLP / SQL / API)?	Wrong listener = silent no-ops
3	Outbound contract(s) (file / SQL / EIP / HTTP) and success criteria ?	Dual-write needs an explicit model
4	Validation level (none / heuristic / SNIP / HL7 structural / partner ACK)?	Don't claim compliance without gates
5	Identity & idempotency keys?	Dedup / retries
6	Happy-path sample(s) available?	Seed data drives the first green path
7	Demo vs production labeling — is this demo-only?	Secrets, SA, PHI mounts
8	Ports free on this machine?	Avoid collisions with other Sandbox demos

Port allocation (default pattern): pick an unused triple near existing demos:

Role	Existing examples
SQL Server host	14335 (EDI), 14336 (HL7)
EIP host	8093 (EDI), 8096 (HL7)
Web UI host	8095 (EDI), 8097 (HL7)

Document chosen ports in `README.md` before `compose up`.

1.1 LAN access (required when spinning up a Web UI)

Whenever an interface Web UI is brought up for local demo / review (`docker compose up`, smoke test, or “spin it up”), agents **must** expose it on the machine’s **192.x LAN address** as well as `localhost`, so it can be opened from phones, tablets, and other devices on the same network.

Required every time:

1. Detect the host’s primary IPv4 on a `192.*` interface (this Sandbox’s usual address is `192.168.68.52` on `en0`; re-detect — do not hardcode forever if DHCP changes).
2. Publish the Web UI with Docker host port mapping (already binds `0.0.0.0` by default) — do **not** bind only to `127.0.0.1`.
3. Set `compose env LAN_HINT=http://<192.x.x.x>:<webui-port>/` so the UI nav shows the LAN link.
4. List **both** URLs in `README.md` and in the agent summary when the stack is running:
 - Local: `http://localhost:<port>/`
 - LAN: `http://192.x.x.x:<port>/`
5. Prefer `Flask/Web UI host=0.0.0.0` (already the demo pattern).

Anti-pattern: shipping or announcing only `localhost / 127.0.0.1` when the user needs to review from another device.

2. Standard directory scaffold

Create under `Clients/Demos/<slug>/` (or `Clients/<Client>/<slug>/` for client work):

```

<interface>/
  README.md
  DESIGN.md                                # filled from §8 template
  docker-compose.yml
  sql/
    01_init.sql                            # idempotent preferred; document if destructive
  samples/                                # inbound fixtures (named clearly)
  input/                                  # runtime drop dirs (gitkeep / .gitkeep)
  output/                                # artifacts (edi, snip, clearinghouse, kickout, archive...)
  eip-root/                              # diagram / V2 source of truth for route viewer
    interfaces/<App>/routes/<N - Name>/
      route.xml
      route.v2.xml                        # generate/convert; don't hand-edit unless required
      modules/*.xml
  pilotfish/
    demo-eip-root/                        # what the container actually mounts at runtime
    Dockerfile                            # FROM pilotfish-eip:23R1 (+ JDBC if needed)
    demo-entrypoint.sh
    conf/environment-settings.conf
  webui/                                  # Flask (or existing demo UI pattern)
    app.py
    templates/
    static/
      route-viewer/                      # copy from an existing demo; keep in sync
  tools/
    convert_routes_to_v2.py               # if needed
    export_route_diagrams.py              # PDF export with --config → documents/
  documents/                              # required deliverable: route design PDF(s)
    <Interface>_V2_Route_Diagrams.pdf
  logs/                                  # gitignored runtime logs if bind-mounted

```

Source of truth: runtime behavior = pilotfish/demo-eip-root.

Diagrams / Routes tab = eip-root (or a documented sync job).

Route design PDF = documents/ (generated; do not leave only under output/route-diagrams).

If runtime and diagrams diverge, fix or document the divergence in DESIGN.md — never leave silent drift.

Reference implementations:

- Clients/Demos/edi-837-snip-sqlserver/ — SQL poll → EIP handoff → fork → EDI + SNIP
- Clients/Demos/hl7-healthcare-automation/ — directory listen → validate → router fan-out → SQL + file

3. Route construction pattern

3.1 Canonical stage chain (V1 runtime)

Author this chain in **V1** route.xml (what pilotfish-eip:23R1 runs):

Listener

- Source processors (normalize / validate / map / snapshot files)
- FormatProfile (relay | XPath fork when true multi-message)
- XPathRoutingModule (Conditional Router)
- Target processors (pre-transport map)
- Transport(s) # Directory | DatabaseSql | EIP | HTTP ...

Then run tools/convert_routes_to_v2.py (or demo equivalent) so route.v2.xml stays aligned for the Routes tab / PDF. Do **not** hand-maintain V2 as the only copy of runtime truth.

3.2 Naming

- Routes: 1 – <Verb Object>, 2 – ... (same spirit as existing demos)
- Modules: stable UUIDs once created; don't regenerate casually (breaks V2 links)
- Env props: `$$$sqlserver.*`, `$$$*_DIRECTORY` in `environment-settings.conf`
- Route viewer / PDF: resolve `$$$NAME`
- Web UI **XSLT** tab: list/view every `.xsl` / `.xslt` under `routes/` (hide the tab when none exist). API: `/api/v2/xslt`, `/api/v2/xslt/content?path=` from that conf for display; **mask** names matching `password/secret/token/apiKey/privateKey` (show `.....`). Directories, drivers, URLs, usernames remain visible.

3.3 Must-design decisions (write into DESIGN.md)

1. When is business state committed?

Never mark work terminal (PROCESSED / archived-as-success) before all required side effects succeed — or document that the demo deliberately does claim-before-complete and list it under Risks.

2. Dual / multi write:

If router fans out to SQL **and** files (or multiple files), define order + compensation (outbox, or “file then SQL”, or accept demo risk explicitly).

3. Fork vs snapshot:

A stage named “Split” / “Batch” must either truly fork (XPathForkingModule / equivalent) or be renamed (e.g. “Batch Flag Snapshot”). No fake splits.

4. Validation gate:

If SNIP / HL7 / XSLT “validation” runs with `StopOnError=false` and always continues, say so. Prefer fail-closed kickout + non-success status for anything marketed as validation.

5. Validate the wire artifact:

Don't SNIP / schema-validate a pre-transform or stripped XML if the partner receives the post-transform payload.

6. Batching / back-pressure:

SQL listeners: bounded TOP (@n) + locking hints when claiming rows.

Triggerable / forked routes: throttling when heavy processors (SNIP) sit downstream.

7. Idempotency:

Unique business keys in SQL; versioned or non-overwrite filenames for audit outputs.

8. Poison / kickout:

Explicit fail directory or status; don't Move-to-archive on exception-only paths.

4. Implementation order (agent checklist)

Work in this order unless the user specifies otherwise:

1. **DESIGN.md** filled from §8 (can be draft; must exist before claiming “done”).
2. **Module selection** via `PilotFish_Documentation` tracker/deep-dives + `PilotFish_V2` (`modules.conf` / module Java). Record chosen FQCNs in DESIGN.md Pipeline table.
3. **SQL init + samples** (if applicable).
4. **environment-settings.conf + compose + Dockerfile** (ports, volumes, heap if SNIP).

5. **Minimal Route 1** V1 route.xml (listener → snapshot/file) green — copy Sandbox skeleton when possible.
6. **Downstream routes / transforms** one stage at a time (still V1 runtime).
7. **Router + transports** with kickout path tested once.
8. **Web UI** inject/submit + status views matching the demo story.
9. **V2 convert** into eip-root + Routes tab / docs.
10. **Route design PDF (required):** with Web UI up, run `python3 tools/export_route_diagrams.py --config changed` and write/copy the PDF into **documents/** at the interface root (see §6).
11. **README.md** (run, ports, smoke commands; link to documents/*.pdf).
12. **Smoke test** (§7) and paste results into the chat (and update DESIGN.md Risks if findings).

Do not refactor unrelated demos. Prefer copying the closest existing demo assembly after modules are chosen from Documentation/V2.

5. Runtime / Docker conventions

- Base image: `pilotfish-eip:23R1` (build from Sandbox root if missing).
 - Bind-mount runtime interfaces and `input/ / output/ / logs/` as other demos do.
 - Heap: if SNIP (or similarly heavy) processors load, set sufficient `CATALINA_OPTS` (EDI demo uses ~6GB max) and document why.
 - Web UI depends on EIP when useful; document cold-start wait (often 60–90s).
 - **LAN:** set `LAN_HINT` to `http://<192.x.x.x>:<webui-port>/` (see §1.1). Host ports must remain reachable on the LAN interface; announce both localhost and LAN URLs after `compose up`.
 - Secrets: demo password may match existing demos for local convenience, but **DESIGN.md** and **README** must say **demo-only** — never invent “production-ready” claims around `sa` + shared passwords.
 - Prefer least-privilege DB users when creating new non-demo client interfaces.
-

6. Web UI, route viewer, and PDF

When the interface has routes (every new Sandbox interface with a route stack):

1. Copy `webui/static/route-viewer/` from a current demo (keep layout + Box config modes).
2. Serve `route.v2.xml` + module XML via `/api/v2/routes...` (mirror existing Flask helpers).
3. Routes tab: route select, layout dropdown, link to docs.
4. Docs / print view and `tools/export_route_diagrams.py` must use the same **Box config** modes:
 - `compact` — names only
 - `changed` — non-default values (**default for docs/PDF**)
 - `all` — all `ModuleConfig` values
5. Export script: `python3 tools/export_route_diagrams.py --config changed`
Screenshot URL must include `mode=docs&bare=1&config=<mode>`.
6. PDF and on-screen docs view should show equivalent config detail (same default).

6.1 Route design PDF → documents/ (required for every new interface)

Any time you **create a new interface** (or add/change routes enough that diagrams should be refreshed), you **must** generate the route design PDF automatically as part of finishing the work — do not wait for the user to ask.

1. Ensure Web UI is up and V2 routes render.
2. Run `python3 tools/export_route_diagrams.py --config changed` (default Box config).
3. Write a **single** PDF under the interface's **documents/** folder (no `_changed` suffix, no second copy):

- documents/<ShortName>_V2_Route_Diagrams.pdf
- 4. PDF layout: **no cover page**; green brand header (and route title) on **every** page.
- 5. Expose the PDF from the Web UI at /documents/route-diagrams.pdf (mount ./documents read-only; Routes tab link **Route design PDF**).
- 6. Mention the PDF path in README.md.

Skipping the PDF requires the user to opt out explicitly; “optional” is not the default.

7. Smoke test (required before “done”)

Minimum bar:

- ☐ docker compose up -d --build comes healthy (or known wait documented)
- ☐ One happy-path sample produces **every** expected side effect (SQL row and/or files)
- ☐ Kickout / fail path exercised once **or** explicitly marked “not implemented” in DESIGN.md
- ☐ Routes tab renders route.v2.xml
- ☐ **Route design PDF** written under documents/ (--config changed)
- ☐ No silent claim of partner-grade validation unless gated

Record: ports, sample used, output paths, and any known failures (e.g. SNIP noise).

8. DESIGN.md template (copy into each interface)

<Interface name> – Design

1. Purpose

<one paragraph>

2. Context / actors

- Sources:
- Destinations:
- Demo vs production:

3. Inbound contract

- Transport:
- Format / envelope:
- Identity fields:
- Samples path:

4. Outbound contract(s)

Destination	Format	Success criterion
-----	-----	-----

5. Pipeline

Stage	Module / mechanism	Notes
-----	-----	-----
Listener		
Processors		
Router		
Transports		

6. State & idempotency

- Status model:

- When state advances:
- Dedup keys:
- Retry / poison:

7. Validation

- What is checked:
- What is NOT checked:
- Does failure block outbound? (yes/no):

8. Dual-write / side effects

- Order of commits:
- Compensation:
- Demo shortcuts (if any):

9. Risks & bottlenecks

Severity	Risk	Why it bites here	Mitigation / accepted?
High			
Med			
Low			

10. Ops

- Ports:
- Volumes:
- Heap / special JVM:
- Dependencies / cold start:

11. Observability

- Logs:
- Kickout dir:
- Transaction / debug tracing: on/off and retention:

12. Open questions

-

Risks checklist (seed §9 from these when applicable)

- Claim-before-complete (status flip before all side effects)
- Dual-write without outbox / compensation
- Validation theater (runs but doesn't gate; wrong artifact validated)
- Fake batch split / unused multi-source folders
- Unbounded poll / no throttling into heavy stages
- Overwrite semantics on audit files
- Missing unique keys → duplicates on resubmit
- Runtime vs eip-root config drift
- Secrets / PHI on permissive mounts
- Partner transport missing (e.g. no MLLP/ACK when story implies it)

9. README.md minimum

Every interface README must include:

1. What it does (bullet flow)
2. Prerequisites (pilotfish-eip:23R1, Docker)
3. docker compose up instructions
4. Ports table

5. **Local and LAN Web UI URLs** (`http://localhost:<port>/` and `http://192.x.x.x:<port>/`)
 6. Smoke / useful commands
 7. Explicit **Demo only** callouts for credentials / PHI / validation limits
-

10. Anti-patterns (do not ship without calling out)

Anti-pattern	Do instead
Name a stage “Validate” / “SNIP” / “Split” that doesn’t gate or fork	Rename or implement the real behavior
UPDATE ... PROCESSED in the same SQL that claims work	Claim → process → terminal status
Router fan-out with retries=1 and no reconcile	Outbox or document as demo risk
Listener FileExtensionRestriction mismatched to story samples	Align contract + samples + UI
Hand-maintained duplicate route trees that disagree	One generator direction; document sync
Claiming “HL7 automation” / “SNIP compliant” from heuristic passes	Accurate language in README + DESIGN
World-writable PHI without demo labeling	Label + restrict for client work
Skipping DESIGN.md because “it’s just a demo”	Short DESIGN.md still required
Announcing only localhost after spin-up	Also set LAN_HINT + list <code>http://192.x.x.x:<port>/</code> (§1.1)
Binding Web UI / publish to 127.0.0.1 only	Keep default 0.0.0.0 host publish so LAN devices can connect

11. Definition of done

An interface construct is done when:

1. DESIGN.md exists and §9 Risks is filled (accepted demo risks allowed if labeled).
 2. Compose stack runs and smoke test passes.
 3. README is enough for a cold start on this machine.
 4. Runtime config and diagrams agree **or** drift is documented.
 5. Web/route viewer present when routes are part of the deliverable.
 6. **Route design PDF** exists under documents/ (generated with `--config changed`).
 7. **LAN URL** is set (LAN_HINT) and both localhost + 192.x URLs are listed when the UI is running (§1.1).
 8. Agent summary lists known limitations in plain language (no compliance theater).
-

12. Quick command pack (adapt ports/paths)

```
cd "Clients/Demos/<slug>"
# detect LAN (example); set LAN_HINT in compose before up
LAN_IP=$(ipconfig getifaddr en0) # macOS; expect 192.*
# ensure docker-compose.yml has: LAN_HINT: "http://192.x.x.x:<webui-port>/" (resolved LAN
IP)
docker compose up -d --build
echo "Local: http://localhost:<webui-port>/"
echo "LAN: http://${LAN_IP}<webui-port>/"
docker compose logs -f pilotfish
# happy path: drop sample or use Web UI
```

```
ls -la output/*
python3 tools/export_route_diagrams.py --config changed
# ensure PDF is under documents/ (exporter may write output/route-diagrams/ - copy if
  needed)
mkdir -p documents && cp -f output/route-diagrams/*Route_Diagrams*.pdf documents/
2>/dev/null || true
ls -la documents/
docker compose down -v # destroys DB volume - warn user first
```